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FoodEx2 maintenance 2023

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Abstract

FoodEx2 is a comprehensive food classification and description system developed and maintained by EFSA to provide a standardized terminology backbone for representing and interpreting samples across the entire food safety risk assessment lifecycle, from data collection across different food and feed safety domains (e.g., monitoring of pesticides residues or biological hazards) to exposure assessment. To keep this terminology and sample coding framework fit-for-purpose, meeting evolving scientific and legislative requirements, a regular maintenance of this system is of utmost importance. Six major updates were carried out so far, with all performed changes documented in annual maintenance reports. This technical report describes the outcome of the seventh maintenance process done in 2023, which contained addition of new terms, inclusion of a new facet hierarchy, creation of two novel implicit attributes, reportability changes of some terms, deprecation of terms and amendments to existing terms including enrichment of implicit facets and changes of the hierarchical tree structure and parent-child relationships.

Key words: food classification, food description, FoodEx2

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Summary

FoodEx2 is a standardised food classification and description system developed by EFSA to improve the comparability and exchange of data coming from different food safety domains within the remit of EFSA. In order to meet this requirement, it needs to be actively maintained, remain up to date as well as flexible to accommodate data collections in different domains. An annual maintenance of the system has taken place since April 2015 when the major revision (FoodEx2 revision 2) was performed.

A maintenance technical group composed of EFSA staff with expertise in different domains (food consumption, chemical contaminants, pesticides residues, etc.) was set-up to evaluate proposals and requests provided by users and stakeholders and take decisions regarding their implementation. The first maintenance was carried out in 2015 followed by six major releases performed between 2016 and 2023.

During the seventh maintenance process performed in 2023, the following major changes have been implemented: 306 new terms have been added to different hierarchies and/or facets to accommodate the needs of a new data collection (notably, the One Health Surveillance) as well as existing data collections (such as the Avian Influenza monitoring programmes and the Chemical monitoring data collection), or requests from other stakeholders in this regard or requests from internal users, forty-seven terms (n=47) have been added based on requests from the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool – GIFT.

Amendments in different sections of the catalogue were implemented during the maintenance of 2023. To accommodate the needs of new data collections such as the One Health surveillance and the Food Additives and Food Flavourings Monitoring, a new facet hierarchy (e.g., F34 Host-sampled facet) and two new novel implicit attributes (e.g., VectorNetCode, ADDFOODCode) were created to the MTX (FoodEx2). Moreover, for the purposes of Food Additives and Food Flavourings Monitoring data collection, the implicit facets are updated for some terms (n=1988) to accommodate the mapping between the FoodEx2 base terms with the legislative classes defined in the Regulation (EC) No 1333/2008 [1] on food additives.

A major revision was undertaken in the Botanicals hierarchy to accommodate maintenance of the EFSA Compendium of Botanicals database. Among these amendments, ninety- seven (n=97) new terms were added to the hierarchy and existing terms were updated (n=193) in terms of extended name, scope notes, parent-child relationship, implicit facet, or other implicit attributes. Most changes (n=187) were related to the update or addition of scientific name/synonyms, which resulted in the update of the scope notes mostly to reflect changes in scientific names. Term extended names were updated for seventy-seven term codes (n=77) and common names were removed from one term (n=1).

An extensive update of the 'Bird (as animal)' section within the F01 – Source facet was also performed in 2023. The facet was restructured and updated to meet the demands of data reporting under the Avian Influenza surveillance programmes in poultry and wild birds. Furthermore, a significant change was implemented to the 'Exposure' hierarchy since it is

[1] Commission Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ L 354, 31.12.2008, p. 16–33.

reconstructed and updated in order to include 22 food groups at the top- level, introducing a separate group for ‘Amphibians, reptiles and terrestrial invertebrates’. The reportability of some terms was updated, as four terms (n=4) are deprecated and forty-two terms (n=42) are dismissed from different hierarchies as found redundant, duplicated or not in line with the data used.

The present report describes the outcome of the seventh maintenance process accomplished in 2023, to keep track of the changes in the system within four minor releases published in 2023 (MTX 14.1, MTX 14.2, MTX 14.3, and MTX 14.4) and the major release published in January 2024 (MTX 15.0).

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1 Introduction

Background and terms of reference as provided by the requestor

In 2011, EFSA released a comprehensive food classification and description system named FoodEx2 (revision 1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out in order to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision 2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), internal and external users should provide to EFSA proposals for implementation and maintenance in order to refine the system and an annual EFSA Maintenance Technical Group (TG) should be set up in order to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system led to the need of keeping the catalogue updated. Following the revision 2 of the system, the FoodEx2 catalogue was revised five times based on EFSA needs and evaluation of the data providers and stakeholders' suggestions. A first maintenance took place in 2015 (EFSA et al., 2016) followed by five major releases performed between 2016-2022. The updated catalogues (MTX 10.0, MTX 11.0, MTX 12.0, MTX 13.0 and MTX 14.0) and the technical reports that summarize all changes performed during the maintenance from 2016 - 2022 were published (EFSA, 2019b, EFSA, 2020, EFSA, 2021b, EFSA, 2022, EFSA,2023).

Moreover, harmonized data collection in the areas of pesticides, contaminant occurrence, veterinary medicinal product residues and food additives (EFSA, 2024), as well as on zoonoses, zoonotic agents, antimicrobial resistance and foodborne outbreaks (EFSA, 2023), is reflected in the maintenance process of the FoodEx2 catalogue. Considering that all the above data domains and a number of other ad-hoc data collections in EFSA use the FoodEx2 catalogue (*e.g.*, food consumption, botanicals *etc.*), there was a need to accommodate different requests. Some changes were performed to enable giving a clear guidance (in terms of FoodEx2 codes) to data providers before the data collection launch date, while others were requested from EFSA staff working in different domains, data providers and network members (hereafter stated as users).

By using the system, both internal and external users and stakeholders provided constructive comments and requests for addition of missing terms and other type of updates in different sections of the catalogue. All these suggestions were considered for the maintenance purposes. A technical group composed of EFSA staff with expertise in different domains carried out the maintenance process, evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the seventh maintenance process undertaken in 2023 to keep track of the changes in the system implemented within four minor releases published in 2023 (MTX 14.1, MTX 14.2, MTX 14.3 and MTX 14.4) and the major release published in January 2024 (MTX 15.0). In order to perform a correct coding, with a reasonable level of detail, this report should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described.

2 Data and Methodologies

2.1 Data

The FoodEx2 catalogue MTX 14.0 - version published in January 2023 was used as a starting point. The suggestions provided during the period between January 2023 and January 2024 by users and stakeholders of the FoodEx2 system were collected through written feedback and used to update the system.

2.2 Methodologies

Each request for addition or changes in the structure was considered and based on expert judgment, the TG decided whether a change was needed, or the request was already covered by the available terms and facet descriptors. All changes were made following the structure and the logic of the system as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

2.2.1 Maintenance Process

To carry out the seventh maintenance of the FoodEx2 catalogue, the TG performed the following steps:

- Evaluation of the requests, comments and proposals from internal and external stakeholders
- Decision on the implementation of the proposals based on the following options: addition of new terms, extension or better specification of the scope of existing terms or identification of the correct way of implementing the suggestion using already existing terms and rules, change of the parent-child relationship of the terms, deprecation and setting of terms as not reportable/dismissed in one hierarchy but still reportable in others.
- Checking for inconsistencies and ambiguities
- Update of the FoodEx2 catalogue
- Summary of the changes for the present technical report.

2.2.2 Tools

The tools used for implementing the maintenance were Microsoft Excel® and the FoodEx2 Catalogue Browser (EFSA, 2019a), a user-friendly tool developed in Java by EFSA staff (Integrated Data former Evidence Management Unit) to help create FoodEx2 codes and to facilitate the management of the MTX catalogue. The scope notes, implicit attributes and facets of some terms were updated using the SAS (SAS Institute Inc., Cary, NC, USA) Enterprise guide 8.3.

3 Maintenance Output

In the following sections, the details on these maintenance operations and rationale behind these changes will be discussed. This overview of the introduced changes is mainly structured along the impacted food safety domains and stakeholders, and legislative changes necessitating these modifications. Horizontal changes (i.e. not requested for a specific data collection) were also performed for maintenance purposes, which affected various aspects (e.g., implicit attributes and/or facet, scope notes etc.) of existing terms from different hierarchies and facets (see Appendix A). For instance, many changes were associated with the modifications performed in the implicit attributes of existing FoodEx2 terms under the section “A026V Fish (meat)” or the term naming and description of terms under the "A04TF Insects (as animal)”.

A detailed table with all type of changes performed during 2023, the hierarchies and facets affected and the rationale behind them, is available in Appendix A.

3.1 Updates for the Avian Influenza surveillance programmes and data collection

FoodEx2 is used for reporting data under the Avian Influenza surveillance programmes in poultry and wild birds, carried out by Member States, according to Regulation (EU) 2016/429 and Commission Delegated Regulation (EU) 2020/689[2]. To meet the data reporting requirements in this domain, an extensive update of the Bird section, 'A057X – Bird (as animal)', was performed in the maintenance process in 2023 due to the addition of new birds requested by data providers to be added to the MTX (FoodEx2). This section required additional structuring in terms of completion of taxonomy and parent-child relationships in order to maintain the consistency of the existing hierarchical structure in FoodEx2 bird section.

An external expert in ornithology was engaged under EFSA's Individual Scientific Advisors (ISA) scheme to provide scientific and technical support in the area of ecosystem analysis. The task involved the assessment of the new birds to be added in the FoodEx2 catalogue requested by data providers and an overall curation of the current taxonomy as well as the collection of detailed information relevant to term names, common names, scientific names, Euring codes etc. Thus, necessary changes and amendments to the current taxonomy were proposed.

To accommodate data providers' requests for the Avian Influenza data collection, n = 106 new bird species have been added and made reportable in the F01 – Source facet during the maintenance process of 2023. The newly added taxonomic entities included three orders (n=3), seventy-nine families (n=79), two genera (n=2), and twenty-one species (n=21). Euring[3] codes were attributed to twenty terms (n=20), reported as necessary for the Avian Influenza Data Collection. Furthermore, two news codes (n=2) related to the testing swabs were added in the F02-Part nature facet (see section 3.7.4).

This systematic revision also motivated changes in existing terms, which were associated with: 1) update or addition of new common and/or scientific names, 2) update of the term extended name, 3) removal of existing common names and/or scientific names to ensure more accurate description of the term names, and 4) alterations to parent-child relationships. Term extended names were updated for six codes (n=6), common names were added or updated for six terms (n=6), while scientific name/synonyms were added or updated for seven terms (n=7). A new parent term was assigned to twenty codes (n=20) mostly due to the addition of missing families in taxonomy (see section 3.7.3). Consequently, scope notes were updated for eleven terms (n=11), mostly to reflect changes in common and/or scientific names (see Appendix A).

3.2 Updates for the One Health surveillance programme and data collection

FoodEx2 is used for reporting data under the One Health Surveillance data collection according to the Commission Implementing Decision of 25.7.2023 amending Implementing Decision C(2022) 317 final of 14 January 2022 on the financing of the Programme for the Union's action in the field of health ('EU4Health Programme') and the adoption of the work programme for 2022[4]. The European Commission (EC) has allocated specific resources for Member States for setting up a coordinated

[2] Commission Delegated Regulation (EU) 2020/689 of 17 December 2019 supplementing Regulation (EU) 2016/429 of the European Parliament and of the Council as regards rules for surveillance, eradication programmes, and disease-free status for certain listed and emerging diseases (Text with EEA relevance). OJ L 174, 3.6.2020, p. 211–340.

[3] Euring is the coordinating organisation for European bird ringing schemes which maintains a databank containing ringing recovery information gathered throughout Europe.

[4] COMMISSION IMPLEMENTING DECISION of 25.7.2023 amending Implementing Decision C(2022) 317 final of 14 January 2022 on the financing of the Programme for the Union's action in the field of health ('EU4Health Programme') and the adoption of the work programme for 2022. Available here: https://health.ec.europa.eu/document/download/18379bcf-ef74-4c7d-9614-dacf3e1d10b9_en?filename=c_2023_5052_1_act_en.pdf

surveillance system under the One Health approach for cross-border pathogens that threaten the Union. The results of this surveillance should be collected by EFSA to perform a risk assessment aiming at identifying One Health zoonotic risks for the EU for which surveillance is needed and allowing for an iterative approach which would allow to review the surveillance priorities. The data for the One Health surveillance are referring to pathogen detection in livestock, wildlife, vectors and environmental samples.

To meet the data reporting requirements in this domain, during the maintenance window of 2023, extensive amendments to different hierarchies and/or facets of the MTX (FoodEx2) were implemented. The majority of these amendments was associated with the addition of new animals as natural sources, followed by the update of existing terms in relation to the scope notes and of parent-child relationships.

Seventy new terms (n=70) were added in different hierarchies of the MTX (FoodEx2). Most terms (n=68) were added to the F01 - Source facet. In particular, thirty-four new terms (n=34) were added to Insects section, 'A04TF - Insects (as animal)', under the category 'Diptera (as animal)' which contains the insects belonging to the taxonomic order Diptera. The newly added taxonomic entities included two families, six genera, twenty-five species and one subspecies. Moreover, twenty-seven new terms (n=27) were added to the category 'Mites and Ticks (as animal)', which contains the animals of the taxonomic subclass Acari, and in particular the orders Ixodida (ticks) and Mesostigmata (mites). The taxonomic entities introduced under this category include two families, seven genera and eighteen species. Seven new terms (n=7) introduced under the Carnivores section 'A0CNK - Carnivores (as animal)', which includes the animals of the taxonomic order Carnivora. This section required additional structuring in terms of completion of taxonomy and parent-child relationships in order to maintain the consistency of the existing hierarchical structure in FoodEx2. New terms were also added under the 'A16PH - Land' and made reportable in the Reporting hierarchy (n=1), and under the 'A0CEG - Non-food animal-related matrices (as part-nature)' (n=1) in the F02 Part-nature facet.

Another amendment to accommodate the needs of data reporting under the One Health surveillance was the creation of a new facet hierarchy to the MTX (FoodEx2) named 'Host-sampled'. This facet provides information on the host-sampled, indicating the host involved in the sampling process, i.e., for samples taken from vectors it indicates the species of the host where the vector was attached at the time of sampling. The set of descriptors (n=1180) introduced in the new facet, F34 Host-sampled, was 'A056H - Mammals (as animal)', 'A057X - Birds (as animal)', 'A0C5Y - Environment' and their children, including the new terms added under the Carnivores and Land section, 'A0CNK - Carnivores (as animal)' and 'A16PH - Land', in the 15.0 MTX (FoodEx2) (see Appendix A). Furthermore, a mapping of the newly introduced vector species added under the sections of 'Insects' and 'Mites and Ticks', with the VectorNetCode for vectors collected under VectorNet project was performed. To accommodate this, a new implicit attribute was created: VectorNetCode. VectorNet is a project of the European Centre for Disease Prevention and Control (ECDC) and EFSA which aims to contribute to improving preparedness and response for vector-borne diseases, following a 'One-Health' approach. It supports the collection of data on vectors and pathogens in vectors related to both animal and human health and hence, a mapping against FoodEx2 codes was considered useful.

Consequently, the new implicit attribute, VectorNetCode, was assigned to forty-four new terms (n=44) in the F01 - Source facet where a direct mapping was found. Considering the amendments in the F01- Source facet, 2 scope notes were updated to better specify the scope of the terms (see Appendix A) and 1 parent-child relationship was updated due to the updated taxonomic structure established in the 'Carnivores' section (see Table 4 - Update of the F01 source facet hierarchy during the maintenance in 2023).

3.3 Updates for the Compendium of Botanicals data collection

EFSA's Compendium of botanicals is a database of botanicals that are reported to contain naturally occurring substances of possible concern for human health when present in food. The compendium is intended to facilitate hazard identification by providing information on composition, toxicity and reported adverse effects for more than 2,500 plants and 600 molecules.

During the maintenance window of 2023, an extensive revision was made in the Botanicals hierarchy with the support of EFSA's contractors, to facilitate the Botanicals data collection. Changes on 295 terms were implemented in the Botanicals hierarchy, including addition of new terms, deprecation or dismission of existing terms as well as changes in term extended name, scope notes, parent-child relationship and in implicit attributes.

Due to this revision, ninety- seven new plant species (n = 97) have been added and made reportable in the Botanicals hierarchy (see Appendix A). The newly added taxonomic entities were added under eighteen different families: Amaryllidaceae, Araliaceae, Apocynaceae, Amaranthaceae, Asparagaceae, Araceae, Ericaceae, Euphorbiaceae, Fabaceae, Malvaceae, Myrtaceae, Ophioglossaceae, Orobanchaceae, Rosaceae, Passifloraceae, Papaveraceae, Tamaricaceae, and Thymelaeaceae. Most newly added terms were complemented with their relevant EPPO (European and Mediterranean Plant Protection Organization) [5] codes when available, a necessary implicit attribute related to the existing internationally agreed terminologies. In addition to this, three existing terms (n=3), previously present only in the F01- Source facet, were added (made reportable) in the Botanicals hierarchy, as well (see Table 1 - List of existing FoodEx2 terms that were made reportable in Botanicals hierarchy).

Table 1 - List of existing FoodEx2 terms that were made reportable in Botanicals hierarchy

FoodEx2 Code	Term Extended Name	Botanicals Parent Code
A18ZK	Chaya (as plant)	A138T
A05GD	Nightshade, black (as plant)	A13DN
A0EDD	Lemon thyme (as plant)	A139Q

The revision also motivated changes in existing terms, which were associated with: 1) update/addition of common names and scientific names, 2) update of the term extended name, 3) removal of existing common names and/or scientific names to ensure more accurate description of the term names, and 4) alterations to the order in which scientific names were originally presented. Most changes (n=187) were related to the update or addition of scientific name/synonyms, which resulted in the update of the scope notes mostly to reflect changes in scientific names. Term extended names were updated for seventy-seven term codes (n=77) and common names were removed from one term (n=1). Hence, the modifications of the scientific names performed in Botanicals hierarchy, also led to the update of the scientific names and/or scope notes for four terms (n=4) that are reportable in Exposure and other hierarchies or facets (see Appendix A).

Four terms (n=4) were deprecated, and one term (n=1) was dismissed from the Botanicals hierarchy since they were found to be duplicated and therefore are not available in the MTX 15.0 release (see section 3.8). Consequently, updates in terms of the scientific names and scope notes were implemented to the existing terms that replaced the deprecated or dismissed terms. Furthermore, the parent-child relationship of four existing terms (n=4) were revised due to the deprecation of the 'A139V Leguminosae - family (as plant)'. Hence, these terms were relocated under 'A138V Fabaceae - family (as plant)' due to the deprecation of its previous parent term

[5] EPPO Global Database provide detailed information for more than 1700 pest species that are of regulatory interest (EPPO and EU listed pests, as well as pests regulated in other parts of the world).

(see Table 2 - Updates in parent-child relationship of existing FoodEx2 terms due to the deprecation of the family 'Leguminosae').

Table 2 – Updates in parent-child relationship of existing FoodEx2 terms due to the deprecation of the family 'Leguminosae'

FoodEx2 Code	Term Extended Name	Botanicals Parent Code Old	Botanicals Parent Code New
A05HN	Cowpea (as plant)	A139V	A138V
A05HP	Garden pea (as plant)	A139V	A138V
A13VT	Bowdichia virgilioides Kunth (as plant)	A139V	A138V
A161V	Vicia sativa subsp. nigra (L.) Ehrh. (as plant)	A139V	A138V

3.4 Updates for the new Food Additives and Food Flavourings Monitoring data collections

In accordance with Regulation (EC) No 1333/2008[6] and Regulation (EC) No 1334/2008[7] the consumption and use of food additives and food flavourings shall be monitored, and data shall be gathered to assess that their use does not raise a health concern. Following a new EC Mandate[8], EFSA should collect data from the Member States, starting with a pilot in 2025, for monitoring the occurrence (analytical results via the Chemical Monitoring data collection), and usage levels (new data collection) of food additives and flavourings. For this purpose, an update on the implicit facets of FoodEx2 base terms as well as the creation of a novel implicit attribute were implemented in the 2023 maintenance window.

Specifically, a mapping between FoodEx2 base terms with the F33- Legislative - classes defined in the Regulation (EC) No 1333/2008 on food additives was applied in the MTX (FoodEx2). The use of the regulatory facet descriptor easily allows domain-specific information to be provided, while still defining the food group in the standard way for the purpose of information storage and retrieval from databases. Hence, the F33 - Legislative-classes facet defined in the Regulation (EC) No 1333/2008 on food additives was implicitly assigned to a total number of 1988 FoodEx2 base terms which are reportable in different hierarchies, including the 'Reporting' and 'Exposure' hierarchies. These changes are presented in the Appendix A.

Another amendment related to the F33 Legislative-classes facet hierarchy was the mapping of the F33 - Legislative classes defined in the Regulation (EC) No 1333/2008 on food additives present in MTX (FoodEx2) with the codes derived from the EFSA's 'ADDFOOD' catalogue, that includes the list of food groups for food additives in the Regulation (EC) No 1333/2008. To accommodate this mapping, a new implicit attribute was created and named as 'ADDFOODCode'. As a result of this activity, the F33 - Legislative-classes facet defined in the Regulation (EC) No 1333/2008 on food additives (n=153) were complemented with the new implicit attribute 'ADDFOODCode', when a direct mapping was found.

[6] Commission Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ L 354, 31.12.2008, p. 16–33.
[7] Regulation (EC) No 1334/2008 of the European Parliament and of the Council of 16 December 2008 on flavourings and certain food ingredients with flavouring properties for use in and on foods and amending Council Regulation (EEC) No 1601/91, Regulations (EC) No 2232/96 and (EC) No 110/2008 and Directive 2000/13/EC. OJ L 354, 31.12.2008, p. 34–50.
[8] Request for scientific and technical assistance in the collection and reporting of the data obtained by the Member States in accordance with Article 27 of Regulation (EC) No 1333/2008 and Article 20 of Regulation (EC) No 1334/2008. Link to the mandate: link here: https://open.efsa.europa.eu/questions/EFSA-Q-2023-00028

3.5 Updates for the Pesticides data collection

The catalogue was also revised for the purposes of the Chemical Monitoring data collection, and in line with Annex I of the Regulation (EC) No 396/2005 on maximum residue levels of pesticides [9]. Twenty- one (n=21) terms received an update of their implicit attribute "matrixCode", which associates a FoodEx2 term with its corresponding legislative category specifying its applicable Maximum Residue Limit (MRL). During the maintenance window of 2023, the matrix code for twenty terms (n=20) have been updated and/or added and one matrix code (n=1) has been removed. Most of the updates (n=15) were implemented to FoodEx2 base terms relevant to the 'Animal carcase'. Table 3 summarizes the updates of the matrix codes as part of the maintenance process of 2023. Furthermore, the term name of two codes (n=2) and the common name of one code (n=1) were updated. In particular, the description of the term 'A16GV Camellia sinensis (tea) fresh leaves' was modified to 'A16GV Camellia sinensis (tea) leaves', while the term description of the code 'A00MB Radish leaves (including radish tops)' was updated to 'A00MB Radish leaves'. Furthermore, an update on the common names of the term 'A010J Topee tambu' was performed since the previously assigned name 'Topinambour' was found to be misleading. Therefore, their scope notes were amended in order to reflect these changes and provide more accurate description.

Table 3 - Updates in the matrix codes of existing FoodEx2 terms in line with the with Annex I of the Regulation (EC) No 396/2005 related to pesticides

FoodEx2 Code	Term Extended Name	matrixCode
A031N	Ostrich eggs	P1030990-003
A049Q	Bovine carcase	P1012990
A04AA	Pig carcase	P1011990
A04AJ	Sheep carcase	P1013990
A04AS	Goat carcase	P1014990
A04BC	Horse carcase	P1015990
A04BL	Asses-mules-hinnies carcase	P1015990
A04BV	Rabbit carcase	P1017990
A04CE	Deer carcase	P1017990
A04CN	Wild boar carcase	P1011990
A04DQ	Chicken carcase	P1016990
A04EA	Turkey carcase	P1016990
A04EJ	Duck carcase	P1016990
A04ES	Goose carcase	P1016990
A04FC	Ratites carcase	P1016990
A0CVM	Nandu eggs	P1030990-002
A0CVN	Emu eggs	P1030990-001
A0DCB	Moringa (dry)	P0401070-001
A0DCH	Soyabeans for consumption (dry)	P0401070-000
A0EYE	Animal carcase	-
A16GV	Camellia sinensis (tea) fresh leaves	P0610000-000

3.6 Terms requested by FAO/WHO

[9] Official Journal of the European Union, Commission Regulation (EU) 2018/62, replacing Annex I to Regulation (EC) No 396/2005 of the European Parliament and of the Council https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32018R0062&from=EN#ntr12-L_2018018EN.01000302-E0012

Forty-seven new terms (n=47) have been added based on requests from the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool (GIFT)[10]. Most of these changes affect the inclusion of novel plant ('as plant') and animal ('as animal') species, in the F01 – Source facet (see Appendix A). Moreover, four existing terms (n=4), present only in the Botanicals hierarchy, were added (made reportable) in the F01- Source facet, as captured in **Error! Reference source not found.**.

If the requested terms were already present in the catalogue under a different name or scientific name, the addition of these terms to the catalogue was not taken into consideration. Instead, the existing term was amended with the common name or synonym of the scientific name, which also led to a better specification of the scope of existing terms (see Appendix A) and coding solutions to the users were provided.

3.7 Amendments in hierarchies and facets

3.7.1. Updates of the Exposure hierarchy

The Exposure hierarchy was designed to facilitate the grouping of food items for the food consumption data collection and the exposure calculations. As described in chapter 3.2.3 of the report on FoodEx2 revision 2 (EFSA, 2015), the structure of the Exposure hierarchy includes 21 groups at the top level (L1). In the maintenance window 2023, a revision of the top-level elements and in particular of the group 'Fish, seafood, amphibians, reptiles and invertebrates' was performed in order to accommodate a request provided by internal users to separate the group for reptiles, amphibians and terrestrial invertebrates in the Exposure hierarchy. The aim of this separation was to avoid grouping results of unrelated food groups and in particular of a commonly consumed food group, such as "Fish and seafood", with foods for which concentration and consumption data are often missing in the exposure assessment ('Amphibians, reptiles, and terrestrial invertebrates').

Hence, a new top-level group (L1) was introduced for the 'Amphibians, reptiles, and terrestrial invertebrates' and thus, concluded to the restructuring of the Exposure hierarchy including 22 groups at the higher level. This revision resulted in further changes in existing terms. Specifically, to accommodate this request, term extended names were updated for two codes. Consequently, in the 15.0 MTX (FoodEx2), the term code 'A026T' corresponds to the grouping of 'Fish and seafood', whereas the term code 'A02KP' represents the grouping of 'Amphibians, reptiles, and terrestrial invertebrates'. Furthermore, scope notes were updated for one term, mostly to reflect changes in the term extended name. The current view of the two new groups is presented in Figure 1.

[10] https://www.fao.org/gift-individual-food-consumption

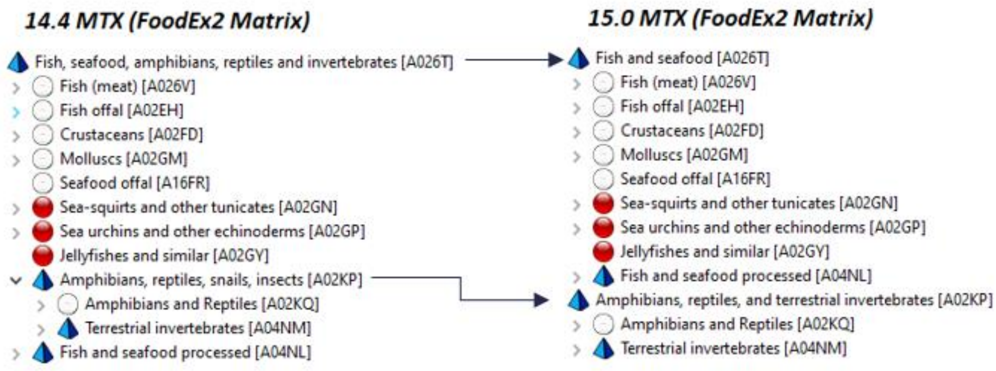

Figure 1: Revision of the "Fish, seafood, amphibians, reptiles and invertebrates" section - changes compared to the previous catalogue version

3.7.2. Updates of the PRIMo hierarchy of FoodEx2

EFSA will launch a new version of the Pesticide Residue Intake Model (PRIMo revision 4) in the first half of 2024. The new tool will be hosted on an online platform with a new user interface and will present various improvements and new features such as, the use of individual-based consumption data in the exposure calculations and the introduction of the PRIMo hierarchy for the food classification. This hierarchy will report all raw primary commodities (RPCs) that can be used in the assessments of PRIMo 4. In order to align with the upcoming version of the tool and to reflect the refinements made on the consumption database, the current PRIMo hierarchy was updated by dismissing forty-one terms (n=41) (see Table 6 **Error! Reference source not found.**3). Moreover, few terms (n=3), which were previously dismissed from the PRIMo hierarchy, were set as reportable again. A new parent term was also assigned to one term (‘A0DCH Soyabeans for consumption (dry)’).

3.7.3 Changes in the F01 – Source facet

F01 – Source facet was amended with a total number of 241 new terms to allow mapping of samples within different data collections and projects. In particular, given the extensive amendment of the Bird section, ‘A057X –Bird (as animal)’, during the maintenance process in 2023, an additional structuring in terms of completion of taxonomy and parent-child relationship was required. Consequently, the addition of new terms requested by data providers led to the update of the parent for existing terms (n=20) in the catalogue. For more details see section 3.1. In addition, the position of the terms “Raccoons (as animal)” and “Lowland paca (as animal)” in the F01-Source facet hierarchy was changed since a more accurate parent-child term relationship was found, due to the new terms added for the purposes of One Health Surveillance data collection and the Prevalence data model (aggregated and sample-based level in SSD2), respectively. For maintenance purposes, a better parent-child term relationship was also observed for the “Turbot (as animal)” due to the modifications performed in the implicit attributes of the codes under the ‘Fish (meat)’ section. Moreover, to accommodate the requests from the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool (GIFT), the applicability of four existing plant species (n=4), which were previously reportable only in Botanicals hierarchy, was updated and therefore were made reportable in F01- Source facet. Table 4 summarizes the updates in terms of the new parent-child relationships and the applicability of terms in the F01- Source facet hierarchy during 2023 maintenance window.

Table 4 - Update of the F01 source facet hierarchy during the maintenance in 2023

FoodEx2 Code	Term Extended Name	Source Parent Code Old	Source Parent Code New
A058E	Helmeted Guineafowl (as animal)	A0CNF	A19EP
A17SK	Podiceps (as animal)	A18JB	A19DC
A17AY	Pied-billed Grebe (as animal)	A18JB	A19DC
A17HB	Western Grebe (as animal)	A18JB	A19DC
A18JC	Tachybaptus (as animal)	A18JB	A19DC
A17NV	Gavia (as animal)	A18HY	A19DE
A17RK	Ciconia (as animal)	A18HQ	A199Z
A17SP	Pelecanus (as animal)	A187S	A19DD
A17YA	Pomarine Skua (as animal)	A17YG	A199Y
A17YB	Arctic Jaeger (as animal)	A17YG	A199Y
A17YC	Long-tailed Skua (as animal)	A17YG	A199Y
A17YD	Great Skua (as animal)	A17YG	A199Y
A17YE	Brown skua (as animal)	A17YG	A199Y
A17YF	South Polar Skua (as animal)	A17YG	A199Y
A183B	Leaf warblers (as animal)	A0CDR	A19BQ
A183E	Kinglets (as animal)	A0CDR	A19CC
A18JL	Swifts (as animal)	A057X	A19DA
A18VH	Speckled Mousebird (as animal)	A18VG	A19CX
A18VK	Violet Turaco (as animal)	A18VJ	A19AD
A18VL	Guinea Turaco (as animal)	A18VJ	A19AD
A18BV	Lowland paca (as animal)	A0CNS	A19DQ
A16DS	Raccoons (as animal)	A0CNK	A198J
A0XZE	Turbot (as animal)	A0XZD	A0XZC
A13TP	Bauhinia variegata L. (as plant)		A16MN
A14ZP	Lawsonia inermis L. (as plant)		A065G
A153G	Madhuca longifolia (J.Koenig ex L.) J.F.Macbr. (as plant)		A05RS
A15RC	Solanum americanum Mill. (as plant)		A16MN

3.7.4 Changes in the F02 –Part-nature facet

F02 –Part nature facet was amended with four new terms (n=4) to allow mapping of samples within different data collections and projects. As described in section 3.1, FoodEx2 is used for reporting data under the Avian Influenza surveillance programmes in poultry and wild birds. To allow an accurate description of the samples taken, there was a need to add two terms related to the testing swabs and in particular ‘A194T Choana swab (as part-nature)’ and ‘A194V Cloacal swab (as part-nature)’, under the code ‘A0CEG Non-food animal-related matrices (as part-nature)’. A new facet descriptor for the 'A197L Embryo (as part-nature)’ under the code ‘A0CEG Non-food animal-related matrices (as part-nature)’ was also added for the purposes of the Prevalence data model (aggregated and sample-based level in SSD2). Furthermore, to allow adequate description of the environmental samples under the One Health Surveillance data collection, there was a need to add a new code for ‘A198B Effluent water (as part-nature)’ under the code ‘A0CEG Non-food animal-related matrices (as part-nature)’.

3.7.5. Changes in the F21 – Production-method facet

To accommodate the needs for zoonoses monitoring activities, specifically for the reporting of *Trichinella*[11] testing of pigs, Facet F21 – Production method was refined during the maintenance window of 2023, where a new term 'A18YP Other production method' was added to the MTX (FoodEx2) in order to refer to any other production method not covered by already existing descriptors.

3.7.6. Creation of new F34 – Host-sampled

To accommodate the needs of data reporting under the One Health surveillance, a new F34-Host-sampled facet hierarchy was created. The new facet provides information on the host-sampled, indicating the host involved in the sampling process. A total number of 1180 terms were set as facet descriptors. For more details and background, see section 3.2 on the One Health surveillance data collection.

[11] EFSA (European Food Safety Authority), 2022. Scientific Network for Zoonoses Monitoring Data Minutes of the 40th meeting. Available online: https://www.efsa.europa.eu/sites/default/files/2022-10/221013_14-m.pdf

3.8 Deprecated and dismissed terms during the maintenance 2023

During the maintenance window of 2023, four terms (n=4) have been deprecated, and therefore, cannot be used anymore (see **Error! Reference source not found.**3). The terms 'A139V Leguminosae - family (as plant)', 'A141X Cheiranthus cheiri L. (as plant)', 'A14SE Holarrhena antidysenterica Wall. ex A.DC. (as plant)' and 'A15FB Polygonum multiflorum Thunb. (as plant)' were deprecated since being a duplicate to and thus replaced by 'A138V Fabaceae - family (as plant)', 'A14HL Erysimum x cheiri (L.) Crantz (as plant)', 'A14SF Holarrhena pubescens Wall. ex G.Don (as plant)' and 'A14LB Reynoutria multiflora (Thunb.) Moldenke (as plant)', respectively.

Error! Reference source not found.3

FoodEx2 Code	Deprecated Term Extended Name	Replaced by
A139V	Leguminosae - family (as plant)	A138V Fabaceae - family (as plant)
A141X	Cheiranthus cheiri L. (as plant)	A14HL Erysimum x cheiri (L.) Crantz (as plant)
A14SE	Holarrhena antidysenterica Wall. ex A.DC. (as plant)	A14SF Holarrhena pubescens Wall. ex G.Don (as plant)
A15FB	Polygonum multiflorum Thunb. (as plant)	A14LB Reynoutria multiflora (Thunb.) Moldenke (as plant)

Table 6 presents the implemented changes in reportability, and in particular the terms which are set as not reportable (dismissed) during the maintenance of 2023. Specifically, one term (n=1) has been dismissed from the Botanicals hierarchy, and several terms (n=41) were dismissed from the PRIMo hierarchy in order to align the current hierarchy with the new version of the PRIMo tool (revision 4) which is currently under development.

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FoodEx2 Code	Term Extended Name	Hierarchy in which the term was dismissed
A00JR	Courgettes	PRIMo
A00JZ	Chayote fruits	PRIMo
A00MJ	Spinaches	PRIMo
A00MN	Purslanes and similar-	PRIMo
A00MV	Oraches leaves	PRIMo
A00VG	Carrageen mosses	PRIMo
A00VM	Kombu	PRIMo
A016D	Other seeds (including elsewhere non-listed oilseeds and other seeds)	PRIMo
A01DP	Pears	PRIMo
A01DQ	Nashi pears	PRIMo
A01FX	Elderberries	PRIMo
A01GD	Sorb fruits	PRIMo
A01HJ	Kumquats and similar-	PRIMo
A01HL	Carambolas and similar-	PRIMo
A01JV	Litchis	PRIMo
A01KQ	Longans	PRIMo
A020R	Bovine other slaughtering products	PRIMo
A021A	Pig other slaughtering products	PRIMo
A021H	Sheep other slaughtering products	PRIMo

A021Z	Poultry other slaughtering products	PRIMo
A02LV	Cow milk	PRIMo
A02MD	Water buffalo milk	PRIMo
A031G	Hen eggs	PRIMo
A031H	Duck eggs	PRIMo
A031J	Geese eggs	PRIMo
A031K	Quail eggs	PRIMo
A03JK	Herbal infusion materials from leaves and herbs	PRIMo
A03JS	Roots used for herbal infusions	PRIMo
A04HK	Equine milk	PRIMo
A04HR	Other farmed terrestrial animals muscle	PRIMo
A04HT	Other farmed terrestrial animals liver	PRIMo
A04JL	Mulberries (black and white) and similar-	PRIMo
A04JQ	American persimmons and similar-	PRIMo
A04JV	Breadfruits and similar-	PRIMo
A04JY	Yams and similar-	PRIMo
A0CGG	Borage seeds and similar-	PRIMo
A0CYJ	Tamarind and similar-	PRIMo
A0CZJ	Carobs and similar-	PRIMo
A0DBJ	Cotton seeds and similar-	PRIMo
A0DCP	Other algae	PRIMo
A0DMC	Okra and similar-	PRIMo
A15YV	Thymus serpyllum subsp. serphyllum (as plant)	Botanicals

Conclusion

The present report documents the update of FoodEx2 performed within four minor releases published in 2023 (MTX 14.1, MTX 14.2, MTX 14.3 and MTX 14.4) and the major release published in January 2024 (MTX 15.0), covering requests from internal and external users.

- In total, 353 new terms have been added to the catalogue since the last major release 14.0.
- Throughout the 2023 maintenance window, 377 scope notes have been updated, to keep their term descriptions and references up-to-date with current scientific knowledge and resources as well as legislative definitions.
- Modifications were performed in existing codes from different hierarchies and/or facets which were associated with update/addition of common names (n= 73) and scientific names (n= 242), update of the term extended name (n= 112), and removal of existing common names (n= 6).
- A new facet hierarchy and a new implicit attribute were introduced in accordance with the One Health Surveillance data collection.
- New terms (n=70) were added in different hierarchies and/or facets and changes on already existing terms (n=3) were implemented in accordance with the One Health Surveillance data collection.
- One new implicit attribute was created in accordance with the Food Additives and Foods Flavourings Monitoring and added to the F33 – Legislative-classes facet for food additives.
- Implicit facets were updated for a total number of 1988 terms in accordance with Food Additives and Food Flavourings Monitoring.
- An extensive update of the Botanicals hierarchy was performed, which concluded to the addition of new terms, changes of already existing codes and changes to reportability.
- An update of the top-level elements of the Exposure hierarchy was performed, leading to 22 food groups at the first level.
- Four terms (n=4) are deprecated while forty-two (n=42) terms are dismissed from different hierarchies as found redundant, duplicated or not in not in line with the data used.
- The Birds section is amended to accommodate Avian Influenza data collection and to complete taxonomy of existing species.
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found mostly due to the addition of new terms.
- The applicability of some terms was extended to further hierarchies.

The complete content of the FoodEx2 system is available in Appendix B to this document, as exported spreadsheet MTX_15.0.xlsx.

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Abbreviations

EC	European Commission
ECDC	European Centre for Disease Prevention and Control
EFSA	European Food Safety Authority
EPPO	European and Mediterranean Plant Protection Organization
EU	European Union
FAO	Food and Agriculture Organization
FoodEx2	Food Classification and Description System
GIFT	Global Individual Food consumption data Tool
IDATA	Integrated Data
ISA	Individual Scientific Advisors
MRL	Maximum Residue Limit
MTX	FoodEx2 Matrix hierarchy
PRIMo	Pesticides Residue Intake Model
TG	Technical Group
VMPR	Veterinary Medicinal Product Residue
WHO	World Health Organization

Appendix A – New terms added and changes made during the 2023 maintenance

Appendix B – FoodEx2 catalogue MTX_15.0